Autonomous Driving Benchmarking

Review 1 and Review 2

Project overview

Project type	Reinforcement-learning project with an applied experimental comparison
Research area	Autonomous driving and continuous control
Primary method	Soft Actor-Critic (SAC)
Baselines	Local language-model planner and human driver
Formal project title	*Autonomous Driving Benchmarking: A Seed-Matched Paired Evaluation of Reinforcement Learning, Large Language Models, and Human Baselines*

The central goal is to develop and evaluate the RL policy. LLM and human driving provide comparison baselines.

Motivation

- Autonomous driving requires an agent to balance lane keeping, speed, progress, and safety.
- Reinforcement learning can learn continuous control through repeated interaction and reward feedback.
- Language-model planners use a different decision process and can provide an interpretable baseline.
- Human control provides a practical reference for interpreting simulated-agent performance.
- Comparing agents on different scenarios makes it difficult to tell whether results come from the agent or scenario difficulty.

Motivating idea

Evaluate the agents in one repeatable driving task with shared scenario seeds and common episode measures.

Problem statement

Problem

Develop a continuous-control RL driving policy and assess its behavior relative to a local LLM planner and a human driver under matched scenario conditions.

Main research question

How does a trained SAC policy compare with a local LLM planner and a human driver on the same seeded driving scenarios?

Comparison dimensions

- Safety: collisions and road departures
- Driving quality: lane and speed tracking, control smoothness
- Task performance: route progress, completion, and time

Objectives and scope

1. Formulate driving as a continuous-control reinforcement-learning task.
2. Train a SAC policy to maintain lane position, track target speed, and make safe progress.
3. Define LLM and human baselines that use the same driving task.
4. Record consistent state, action, and episode telemetry.
5. Compare agents on shared seeds using relevant measures.

Scope: controlled simulated driving and agent comparison. Results are specific to the task and should not be presented as evidence of real-road driving ability.

Literature review: reinforcement learning for control

Soft Actor-Critic (SAC)

Haarnoja et al. (2018) introduce an off-policy maximum-entropy RL method for continuous action spaces.

- A replay buffer allows SAC to reuse past transitions.
- The entropy term encourages exploration while optimizing expected reward.
- Continuous actions fit steering and throttle/brake control.
- This project uses SAC as its primary learning method.

Design implication: evaluate both task reward and driving measures such as safety, lane tracking, and speed error.

Literature review: language models in planning

Work	Main idea	Relevance to this project
SayCan (Ahn et al., 2022)	Combines language-model planning with grounded low-level skills	Motivates separating semantic decisions from vehicle control
GPT-Driver (Mao et al., 2023)	Frames motion planning as language generation	Connects language models with driving decisions
LanguageMPC (Sha et al., 2023)	Integrates language-model decisions with model-predictive control	Motivates an LLM planner paired with a deterministic controller

Project baseline: the LLM proposes a high-level maneuver; a low-level controller converts it to continuous actions. Its output and fallback status must be logged.

Literature review: human baselines and fair comparison

Human driving

- CARLA (Dosovitskiy et al., 2017) established a widely used autonomous-driving simulation platform.
- A live human baseline can provide a direct reference within the same task.
- Human trials need consistent instructions and scenario seeds.

Paired evaluation

- Demšar (2006) discusses statistical comparisons across multiple datasets.
- Matching scenario seeds controls for differences in scenario difficulty.
- Analyze within-seed differences and report effect sizes and uncertainty.

These works inform the study design; they do not by themselves establish the project's novelty.

Research gap and proposed contribution

Working research gap

The project asks whether a small local LLM planner, a learned SAC policy, and a live human baseline can be compared fairly when they encounter the same seeded task and share evaluation measures.

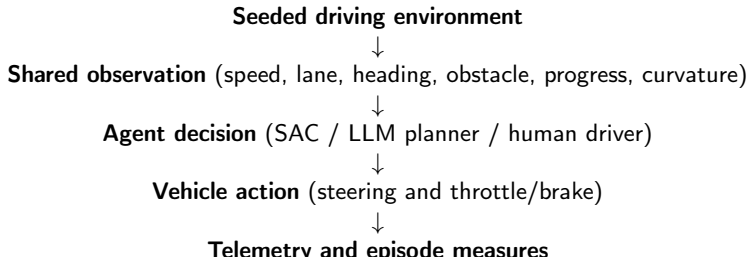
Proposed contribution

- A repeatable seeded driving task and common telemetry format
- SAC as the primary learned continuous-control agent
- LLM and human baselines evaluated under matched scenarios
- Per-seed analysis of safety, tracking, smoothness, and task completion

Review 1 summary

- The project is an RL study centered on training a SAC driving policy.
- The comparison baselines are a local LLM planner and a human driver.
- Shared scenario seeds and common measures are central to a fair comparison.
- The literature motivates SAC, hierarchical language-model planning, human baselines, and paired evaluation.
- The methodology uses SAC as the main learned controller and evaluates it alongside LLM and human baselines.

Review 2: system architecture



- SAC produces continuous actions at each 20 Hz simulation step.
- The LLM proposes a high-level maneuver at a configured 1 Hz; a deterministic controller converts it to continuous controls.
- The human baseline uses keyboard control in the same task.

Review 2: SAC model and learning objective

Primary method

Soft Actor-Critic is an off-policy, maximum-entropy algorithm for continuous actions. Its replay buffer reuses collected transitions during training.

Observation

- Speed and target speed
- Lane deviation and heading error
- Obstacle distance and route progress
- Road curvature

Action

- Steering in $[-1, 1]$
- Throttle/brake in $[-1, 1]$
- Continuous control at each environment step

Configured model: MLP policy, learning rate 3×10^{-4}, replay buffer 50,000, batch size 256, discount factor $\gamma = 0.99$.

Review 2: reward design

The reward combines the task's driving objectives:

$$R = w_p R_{progress} + w_l R_{lane} + w_v R_{speed} + w_s R_{safety} + w_t R_{task} + w_c R_{control}$$

Component	Environment signal	Intended behavior
Progress	Speed and route progress	Move toward the destination
Lane keeping	Lane deviation and heading error	Stay near the lane center
Speed tracking	Error from target speed	Maintain the desired speed
Safety	Collision and road-departure penalties	Avoid unsafe termination
Task completion	Destination reward	Complete the route
Control	Steering penalty	Avoid excessive steering

Review 2: experimental setup

Driving task

- Lightweight top-down 2D road simulator
- 20 Hz simulation with 40 km/h target speed
- Generated obstacle and route-completion signal
- Gymnasium environment interface

Scenario seeds

- Training: 20 seeds, 1000–1019
- Validation: 10 seeds, 1100–1109
- Held-out test: 20 seeds, 2000–2019
- Use the same held-out seeds for each agent

Keep test scenarios separate from model selection and tuning.

Review 2: data collection and analysis workflow

Stage	Project data or method	Review status
Collection	Per-step JSON telemetry for seed, agent, state, action, termination, and LLM backend	Implemented; pilot volume is small
Feature extraction	Speed, lane deviation, heading error, route progress, steering, and throttle/brake	Implemented
Preprocessing	Validate fields and match episodes by seed before comparing agents	Full-dataset checks remain
EDA	Metric summaries, trajectories, speed profiles, and SAC evaluation curve	Pipeline available; pilot only

Episode measures: collision, road departure, completion, route progress, lane-deviation RMSE, speed-error MAE, steering smoothness, and LLM latency/fallback status.

Review 2: comparison plan and current evidence

Analysis plan

- Match records by scenario seed.
- Compare within-seed outcomes.
- Use paired t-tests or Wilcoxon tests where appropriate.
- Report sample counts, effect sizes, and uncertainty.

Pilot status

- SAC trained for 5,000 steps.
- Validation reward: 886.61 ± 472.52 over five episodes.
- One 80-step matched smoke pair on seed 2003 ended at the time limit without collision.
- Ollama was unavailable; no human run has been collected.

Evidence limit

The pilot is too small and variable to support conclusions about which agent performs best.

Combined review takeaways

- Review 1 defines the RL problem, objectives, literature basis, and comparison question.
- Review 2 specifies the SAC controller, observation and action spaces, reward, and experiment design.
- The study compares SAC with LLM and human baselines under matched scenario conditions.
- The next work is to extend training, complete baseline runs, validate the full telemetry, and analyze held-out scenarios.

References

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